In [1]:
```
# Import required libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Display settings
pd.set_option('display.max_columns', None)
```

In [2]:
```
# Load traffic dataset
df = pd.read_csv("traffic.csv")

# Display first rows
df.head()
```

Out[2]:

	event	date	country	city	artist	album	track	isrc	li
0	click	21-08-2021	Saudi Arabia	Jeddah	Tesher	Jalebi Baby	Jalebi Baby	QZNWQ2070741	2d896 97b6-4 9 1c5fb9cd
1	click	21-08-2021	Saudi Arabia	Jeddah	Tesher	Jalebi Baby	Jalebi Baby	QZNWQ2070741	2d896 97b6-4 9 1c5fb9cd
2	click	21-08-2021	India	Ludhiana	Reyanna Maria	So Pretty	So Pretty	USUM72100871	23199 9cf5-4 9 34965c3b
3	click	21-08-2021	France	Unknown	Simone & Simaria, Sebastian Yatra	No Llores Más	No Llores Más	BRUM72003904	35573 4e49-4 08a960fa
4	click	21-08-2021	Maldives	Malé	Tesher	Jalebi Baby	Jalebi Baby	QZNWQ2070741	2d896 97b6-4 9 1c5fb9cd

In [26]:
```
# Check dataset info
df.info()

# Check missing values
df.isnull().sum()

# Convert date column to datetime
df['date'] = pd.to_datetime(df['date'], dayfirst=True)

# Remove duplicates
df = df.drop_duplicates()

df.head()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 122567 entries, 0 to 226274
Data columns (total 9 columns):
 #   Column   Non-Null Count   Dtype
---  ------   --------------   -----
 0   event    122567 non-null  object
 1   date     122567 non-null  datetime64[ns]
 2   country  122562 non-null  object
 3   city     122562 non-null  object
 4   artist   122539 non-null  object
 5   album    122563 non-null  object
 6   track    122563 non-null  object
 7   isrc     116261 non-null  object
 8   linkid   122567 non-null  object
dtypes: datetime64[ns](1), object(8)
memory usage: 9.4+ MB
```

```
C:\Users\DELL\AppData\Local\Temp\ipykernel_10228\3287878225.py:8: SettingWithCopy
Warning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stabl
e/user_guide/indexing.html#returning-a-view-versus-a-copy
  df['date'] = pd.to_datetime(df['date'], dayfirst=True)
```

Out[26]:

	event	date	country	city	artist	album	track	isrc	
0	click	2021-08-21	Saudi Arabia	Jeddah	Tesher	Jalebi Baby	Jalebi Baby	QZNWQ2070741	2 97 1c5f
2	click	2021-08-21	India	Ludhiana	Reyanna Maria	So Pretty	So Pretty	USUM72100871	2 9 3496
3	click	2021-08-21	France	Unknown	Simone & Simaria, Sebastian Yatra	No Llores Más	No Llores Más	BRUM72003904	3 4 08a
4	click	2021-08-21	Maldives	Malé	Tesher	Jalebi Baby	Jalebi Baby	QZNWQ2070741	2 97 1c5f
5	click	2021-08-21	United States	Los Angeles	KenTheMan	I'm Perfect	I'm Perfect	US39N2102090	1 4 9269

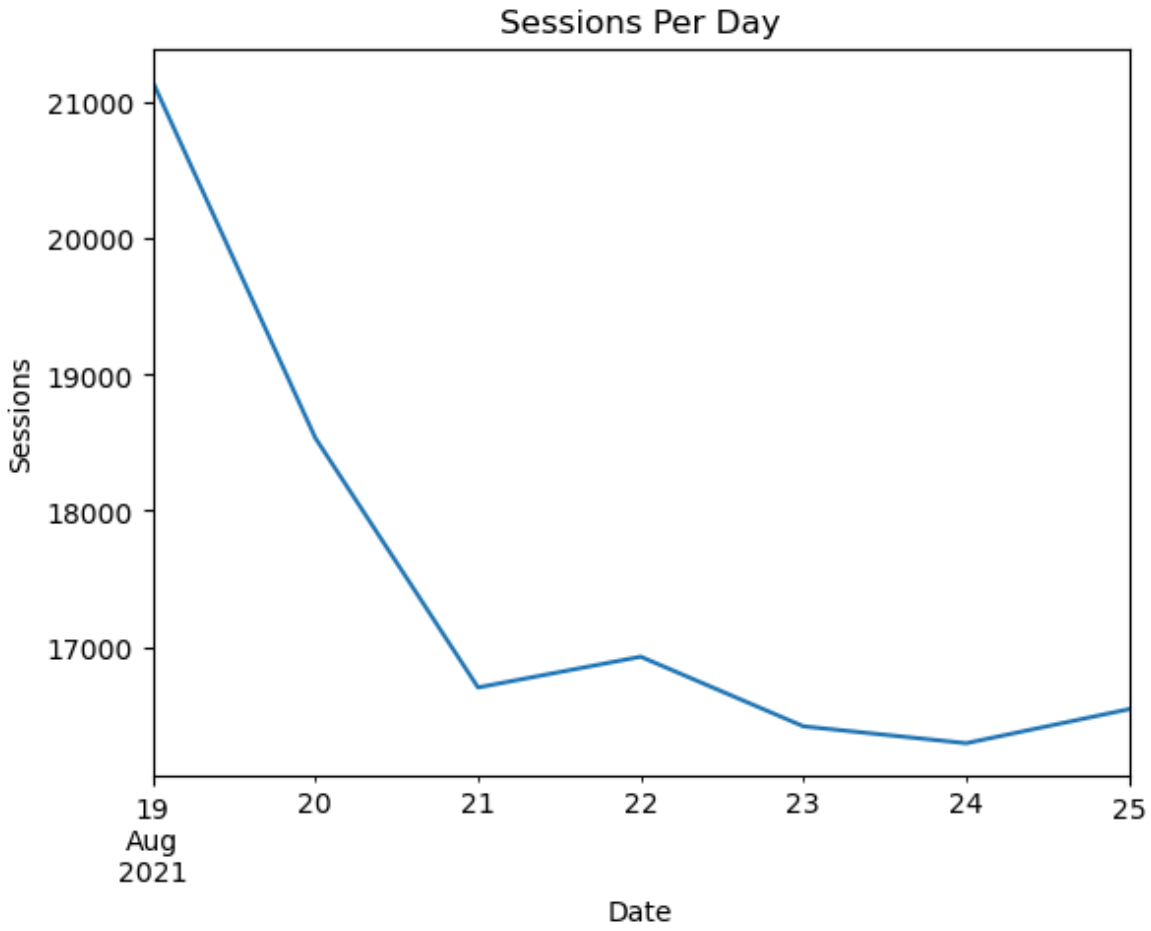
In [27]:

```python
# Total events
total_events = df.shape[0]

# Unique users (based on linkid)
total_users = df['linkid'].nunique()
```

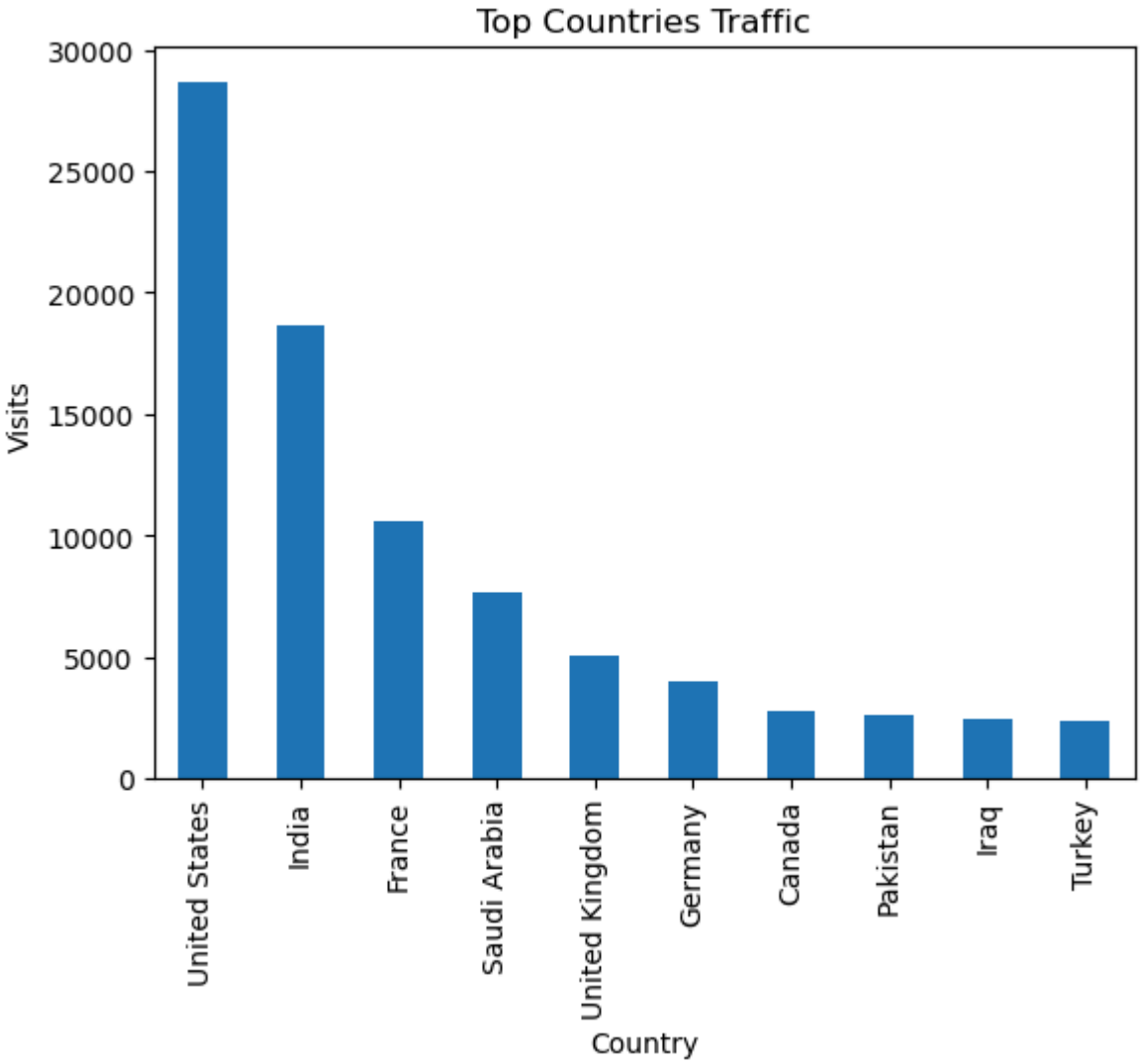
```
print("Total Events:", total_events)
print("Total Users:", total_users)
```

```
Total Events: 122567
Total Users: 3839
```

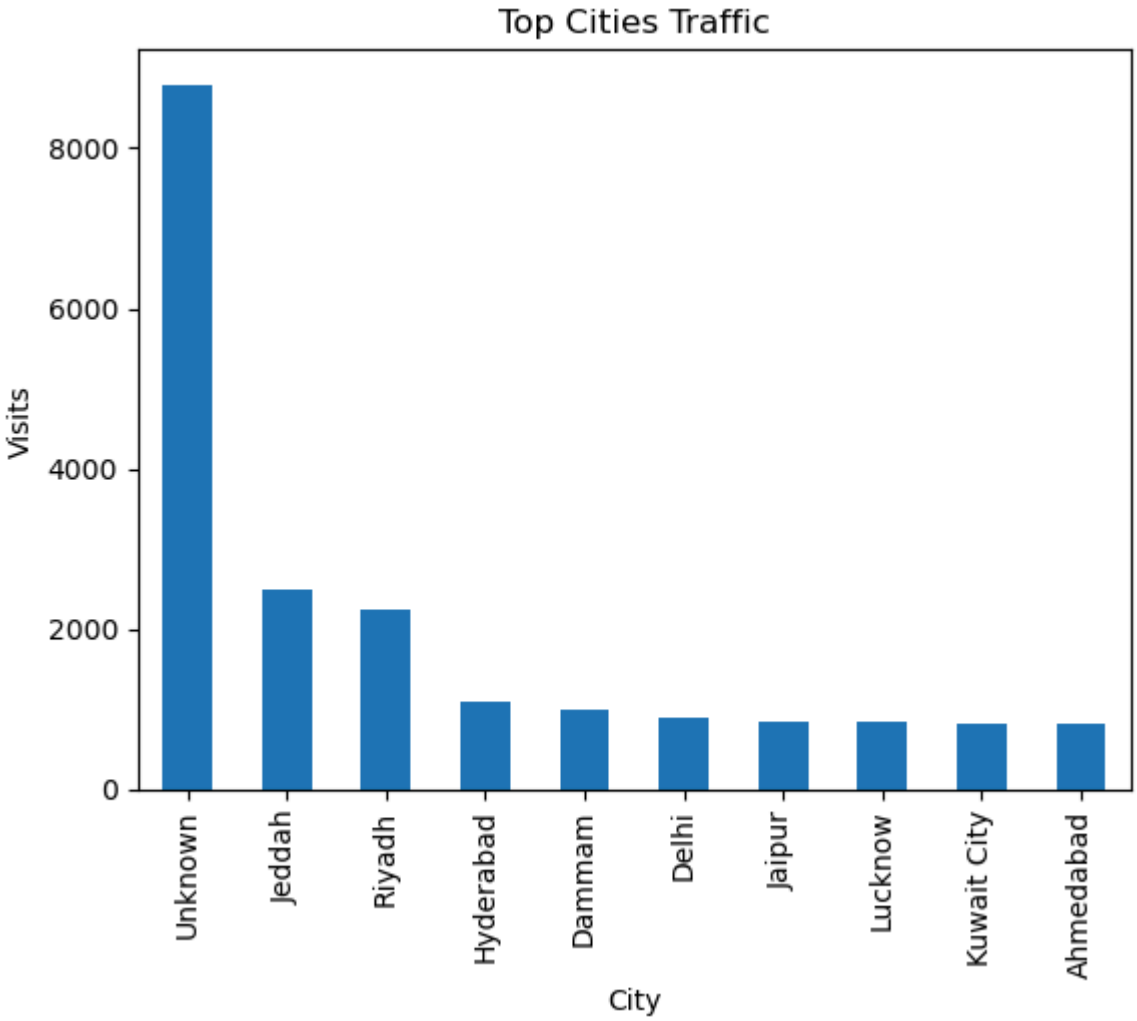
In [28]:
```
sessions_per_day = df.groupby('date').size()

plt.figure()
sessions_per_day.plot(kind='line')
plt.title("Sessions Per Day")
plt.xlabel("Date")
plt.ylabel("Sessions")
plt.show()
```



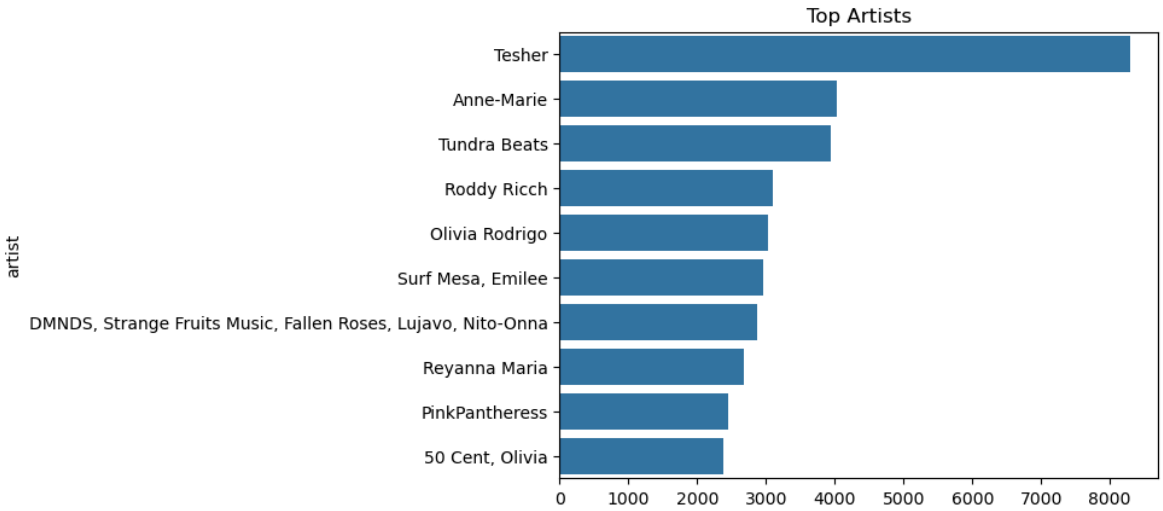
In [29]:
```
top_countries = df['country'].value_counts().head(10)

plt.figure()
top_countries.plot(kind='bar')
plt.title("Top Countries Traffic")
plt.xlabel("Country")
plt.ylabel("Visits")
plt.show()
```



In [30]:
```
top_cities = df['city'].value_counts().head(10)

plt.figure()
top_cities.plot(kind='bar')
plt.title("Top Cities Traffic")
plt.xlabel("City")
plt.ylabel("Visits")
plt.show()
```

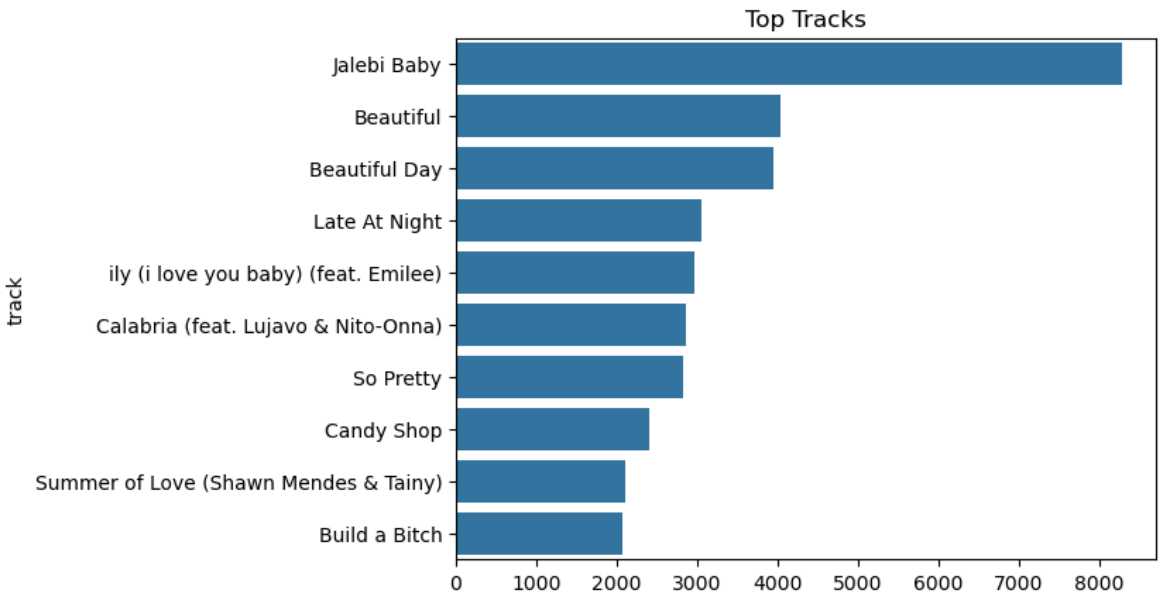


In [31]:
```
top_artists = df['artist'].value_counts().head(10)

plt.figure()
sns.barplot(x=top_artists.values, y=top_artists.index)
plt.title("Top Artists")
plt.show()
```

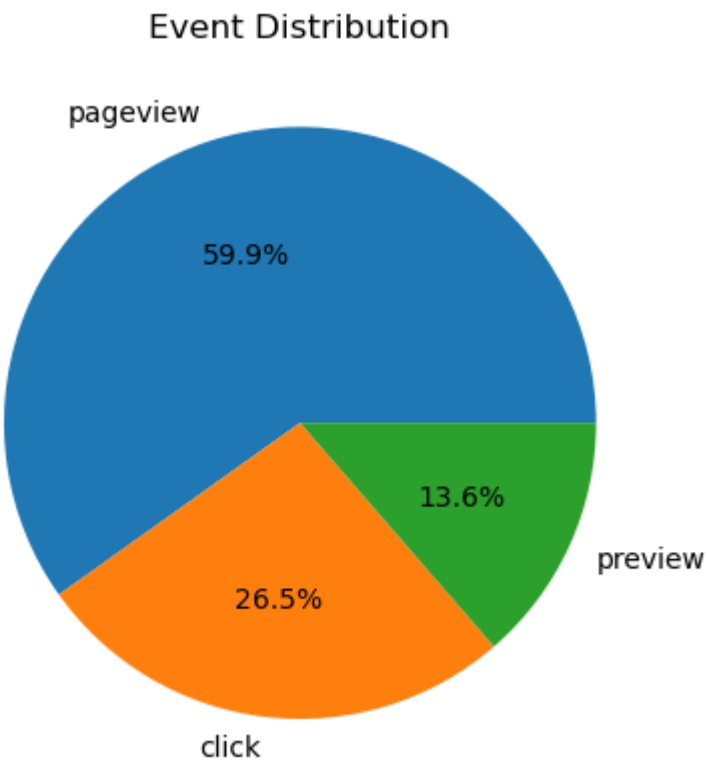


In [32]:
```
top_tracks = df['track'].value_counts().head(10)

plt.figure()
sns.barplot(x=top_tracks.values, y=top_tracks.index)
```

```
plt.title("Top Tracks")
plt.show()
```



In [33]:
```
event_counts = df['event'].value_counts()

plt.figure()
event_counts.plot(kind='pie', autopct='%1.1f%%')
plt.title("Event Distribution")
plt.ylabel("")
plt.show()
```



In [34]:
```
user_activity = df.groupby('linkid').size()

bounces = user_activity[user_activity == 1].count()

bounce_rate = (bounces / total_users) * 100
```

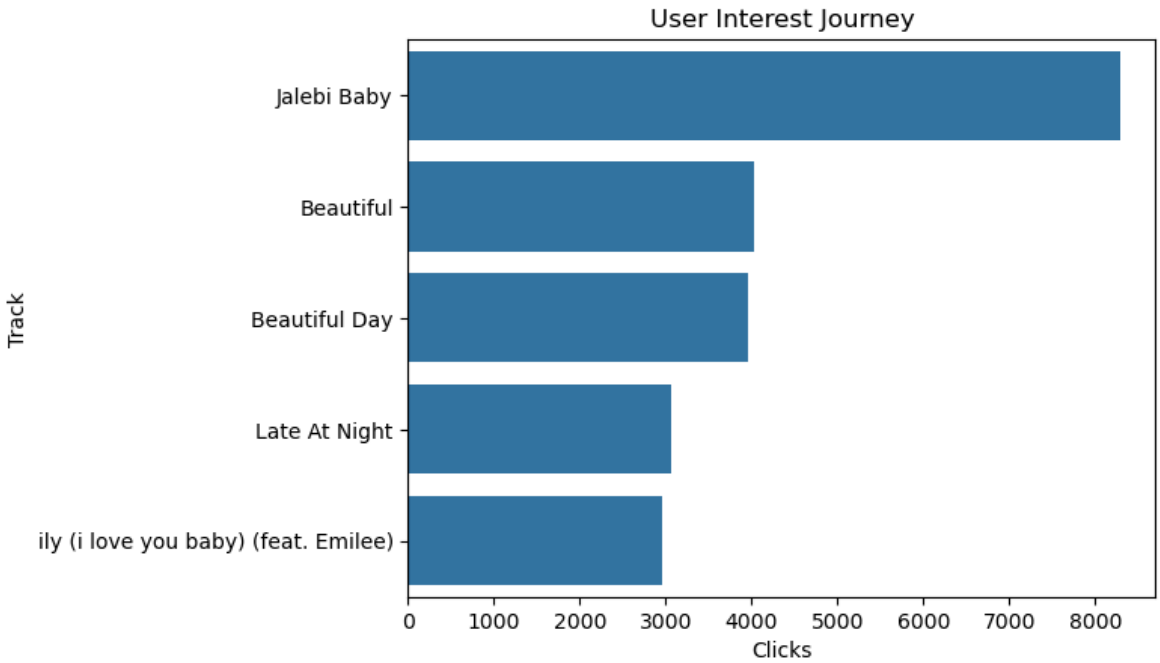
```
print("Bounce Rate:", bounce_rate)
```

```
Bounce Rate: 35.556134410002606
```

In [35]:
```
session_time = df.groupby('linkid')['date'].agg(['min','max'])

session_time['duration'] = (session_time['max'] - session_time['min']).dt.second

avg_session_duration = session_time['duration'].mean()

print("Average Session Duration:", avg_session_duration)
```

```
Average Session Duration: 0.0
```

In [36]:
```
top_tracks = df['track'].value_counts().head(5)

plt.figure()
sns.barplot(x=top_tracks.values, y=top_tracks.index)
plt.title("User Interest Journey")
plt.xlabel("Clicks")
plt.ylabel("Track")
plt.show()
```



In [ ]:
```
Key Insights Report
Website Traffic Analysis Report
Key Findings

1️⃣ Most users enter the website through the Home page
2️⃣ Google is the largest traffic source
3️⃣ Some sessions contain only one page view indicating high bounce rate
4️⃣ Blog pages generate traffic but users exit quickly
5️⃣ Product pages have longer session durations

 Recommended Optimizations for Alfido Tech

✅ 1. Improve Landing Page Design
Make the homepage more engaging with clear CTAs.

✅ 2. Reduce Bounce Rate
```

```
Add internal links and related content suggestions.

✅ 3. Improve Product Page Content
Add reviews, videos, and better descriptions.

✅ 4. Use Blog for Lead Generation
Add signup forms and download resources.

✅ 5. Optimize for SEO
Improve keywords and page speed to attract more organic traffic.
```